Mahmoud Alyosify

Machine Learning Engineer

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SUMMARY

Machine learning engineer who takes a model **from raw data to a working service**: data preparation, training, fair evaluation and deployment with FastAPI, ONNX and AWS. MSc in Artificial Intelligence, Queen's University, Canada (GPA 4.30 / 4.30). Currently an AI Engineer Intern at Injaz Tech.

SKILLS

Machine learning: PyTorch, TensorFlow, scikit-learn, XGBoost, deep learning, computer vision, NLP
Workflow: feature engineering, model evaluation, hyperparameter tuning (Optuna), imbalanced data
Deployment: FastAPI, ONNX, Docker, AWS, Git
Data: Python, SQL, Pandas, NumPy

EXPERIENCE

AI Engineer Intern — Injaz Tech (Injaz Group), Cairo — Oct 2026 – Present

- Built a RAG assistant that answers staff questions from company documents, using **LangChain, LangGraph, FAISS** and **FastAPI**.
- Raised top-1 retrieval accuracy from **75% to 94%** and cut false refusals from 5 in 20 to 1 in 20.

Part-Time Machine Learning Instructor — National Telecommunication Institute (NTI), Egypt — Jul 2025 – Present

- Teach **90-hour machine learning bootcamps** and AI for Business modules in NTI's Summer Internships Program.

Systems Engineer (Military Service) — National Company for Roads Building & Development, Cairo — Feb 2024 – Mar 2025

- Supported the backend systems of Egypt's national toll network, used by **more than half of the country's vehicles**.

Online Instructor — Udemy & YouTube (Arabic courses), Remote — 2020 – Present

- Created Arabic courses on algorithms, discrete mathematics, probability and statistics: **10,000+ learners, 700K+ views**.

PROJECTS

SimCLR-Vision-SSL — self-supervised image model (PyTorch, ONNX, FAISS) — 2026

- Trained an image model without labels (SimCLR), tested **42 augmentation settings** and reached **84.30% top-1 accuracy** with a semi-supervised extension (SupCon).
- Exported the model to **ONNX** and served image search through a **FAISS** index, ready to deploy.

SCRAP — satellite collision risk prediction (Python, scikit-learn) — 2026

- Predicted satellite collision risk using only data available **48+ hours** before closest approach, so the warning arrives in time to act.

Horus-OSINT — fine-tuned LLM on AWS (Llama-3-8B, QLoRA, AWS) — 2026

- Fine-tuned **Meta-Llama-3-8B** on **159,826** global event and terrorism records (GTD, GDELT) and deployed it on **AWS** as an assistant that answers questions in plain language.

EDUCATION

MSc in Artificial Intelligence — Queen's University, Kingston, Canada — Sep 2025 – Oct 2026

- GPA **4.30 / 4.30** · Completed October 2026 · Digilians Presidential Scholarship
- Thesis: MinimalLM — shorter LLM answers that stay complete (supervisor: Dr. Rahatara Ferdousi)

BSc in Computer & Information Science (Bioinformatics) — Assiut University, Egypt — Sep 2019 – Jul 2023

- GPA 3.53 / 4.00 (Very Good with Honours) · Graduation project graded A+

CERTIFICATIONS & AWARDS

IBM AI Developer Professional Certificate (IBM, 2026) · Deep Learning Specialization (DeepLearning.AI, 2026) · Machine Learning Specialization (DeepLearning.AI) · AWS Academy: ML Foundations, ML for NLP, Data Engineering · 1st Place, Smart Cities Hackathon 2022 (77 university teams)

LANGUAGES

Arabic (native) · English (professional working proficiency)